

EXPERIMENT - 5

IPTables Firewall Configuration in Kali Linux

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux virtual machine. Log in to the system.

Step 2: Open Terminal and Check IPTables

Open the terminal and verify that IPTables is installed and available in the system.

Step 3: View Existing Rules

Check the current firewall rules to understand the default configuration before making changes.

Step 4: Configure Firewall Rules

Set default policies to control incoming and outgoing traffic.
Allow necessary services such as localhost communication, SSH, HTTP, and HTTPS.
Block unwanted or suspicious IP addresses.

Step 5: Save and Verify Rules

Save the configured firewall rules and verify that they are applied correctly by reviewing the rule list.

```
(student@kali)-[~]
$ iptables --version
iptables v1.8.11 (nf_tables)

(student@kali)-[~]
$ sudo iptables -L
[sudo] password for student:
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

(student@kali)-[~]
$ sudo iptables -P INPUT DROP

(student@kali)-[~]
$ sudo iptables -P OUTPUT ACCEPT

(student@kali)-[~]
$ sudo iptables -A INPUT -i lo -j ACCEPT

(student@kali)-[~]
$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

(student@kali)-[~]
$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

(student@kali)-[~]
$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

(student@kali)-[~]
$ sudo iptables -A INPUT -s 192.168.1.100 -j DROP

(student@kali)-[~]
$ sudo iptables-save>firewall_rules.txt
```

```

(student@kali)-[~]
$ sudo iptables -L -v -n
Chain INPUT (policy DROP 176 packets, 21154 bytes)
 pkts bytes target     prot opt in     out     source           destination
  0      0 ACCEPT     all  --  lo     *       0.0.0.0/0         0.0.0.0/0
  0      0 ACCEPT     tcp  --  *      *       0.0.0.0/0         0.0.0.0/0
  0      0 ACCEPT     tcp  --  *      *       0.0.0.0/0         0.0.0.0/0
  0      0 ACCEPT     tcp  --  *      *       0.0.0.0/0         0.0.0.0/0
  0      0 DROP      all  --  *      *       192.168.1.100     0.0.0.0/0

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source           destination

Chain OUTPUT (policy ACCEPT 88 packets, 6598 bytes)
 pkts bytes target     prot opt in     out     source           destination

(student@kali)-[~]
$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source           destination
ACCEPT     all  --  anywhere         anywhere
ACCEPT     tcp  --  anywhere         anywhere      tcp dpt:ssh
ACCEPT     tcp  --  anywhere         anywhere      tcp dpt:http
ACCEPT     tcp  --  anywhere         anywhere      tcp dpt:https
DROP       all  --  192.168.1.100     anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source           destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source           destination

(student@kali)-[~]
$ sudo iptables -S
-P INPUT DROP
-P FORWARD ACCEPT
-P OUTPUT ACCEPT
-A INPUT -i lo -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 80 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 443 -j ACCEPT
-A INPUT -s 192.168.1.100/32 -j DROP

```

```
(student@kali)-[~]
$ sudo iptables -L INPUT -v -n
Chain INPUT (policy DROP 316 packets, 37818 bytes)
pkts bytes target     prot opt in     out     source            destination
0      0 ACCEPT     all  --  lo     *       0.0.0.0/0          0.0.0.0/0
0      0 ACCEPT     tcp  --  *      *       0.0.0.0/0          0.0.0.0/0
0      0 ACCEPT     tcp  --  *      *       0.0.0.0/0          0.0.0.0/0
0      0 ACCEPT     tcp  --  *      *       0.0.0.0/0          0.0.0.0/0
0      0 DROP       all  --  *      *       192.168.1.100     0.0.0.0/0

(student@kali)-[~]
$ sudo iptables -L INPUT -n | grep DROP
Chain INPUT (policy DROP)
DROP all -- 192.168.1.100 0.0.0.0/0

(student@kali)-[~]
$ cat firewall_rules.txt
# Generated by iptables-save v1.8.11 (nf_tables) on Tue Feb  3 16:08:28 2026
*filter
:INPUT DROP [150:18953]
:FORWARD ACCEPT [0:0]
:OUTPUT ACCEPT [86:6518]
-A INPUT -i lo -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 80 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 443 -j ACCEPT
-A INPUT -s 192.168.1.100/32 -j DROP
COMMIT
# Completed on Tue Feb  3 16:08:28 2026
```

Result

Various firewall rules were implemented using IPTables, demonstrating how network traffic can be controlled to enhance system security.